

PROJECT

5.6

VPC PEERING

NETWORKING SERIES

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Objective

This project aims to set up multiple VPCs to simulate separate, isolated networks, just like having two cloud neighborhoods. Each VPC is independently managed and can host its own resources securely.

To enable communication between these VPCs, a VPC peering connection is created. This allows private traffic to flow between them without using the public internet, enabling secure, low latency networking across your cloud environments

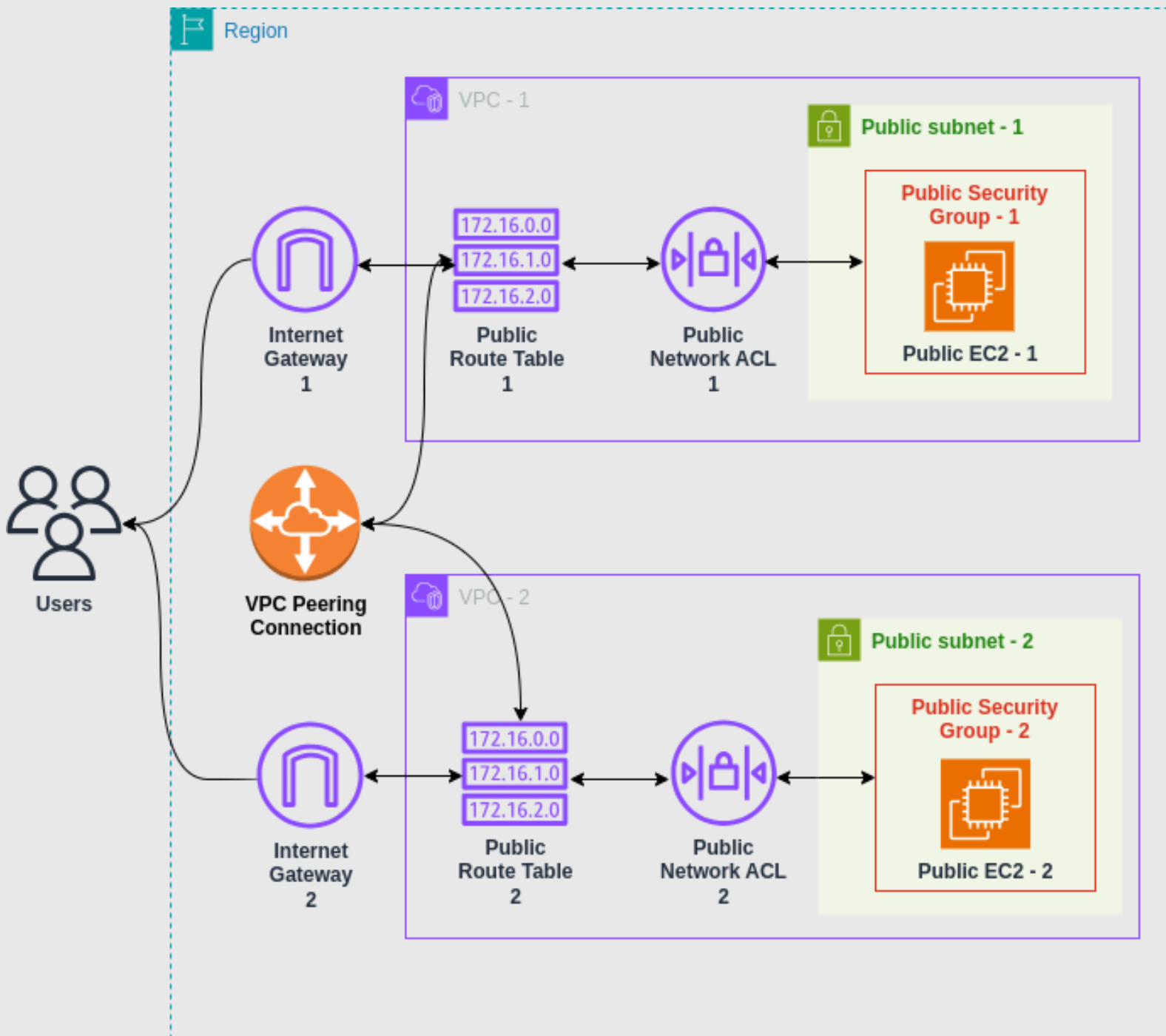
Services

Amazon VPC - Lets you create isolated cloud networks with customizable IP ranges.

VPC Peering - Connects two VPCs privately so they can communicate as if on the same network.

EC2 Instances - Used to test and validate communication between peered VPCs.

Architecture Diagram



The architecture includes two separate VPCs, each with its own subnet and EC2 instance. A VPC peering connection links them, allowing traffic to flow privately between the two networks. Route tables are updated to enable communication, while security groups ensure only allowed traffic passes through.

Steps to Implement

1. Create Two VPCs with EC2 Instances:

Start by creating two VPCs with non-overlapping CIDR blocks (like 10.0.0.0/16 and 20.0.0.0/16). Launch one EC2 instance in each VPC to test connectivity later. Make sure each VPC has its own subnet and internet gateway if needed.

2. Establish a VPC Peering Connection:

Go to the VPC dashboard and create a peering connection from one VPC to the other. Accept the peering request from the second VPC to finalize the connection. This acts like a private bridge between the two VPCs.

3. Update Route Tables for Communication:

Modify the route tables in both VPCs to include routes pointing to each other's CIDR blocks through the peering connection. This allows traffic to flow privately between the VPCs. Without this, they won't know how to reach one another.

4. Test Private Connectivity Between EC2s:

Use the private IPs of the EC2 instances to ping or SSH from one to the other. If the connection works, VPC peering is successfully set up. This confirms that routing and security group settings are correctly configured.

Story Time

VPCs (Two Cloud Cities):

Build two separate cloud kingdoms [VPC 1 and VPC 2] each with their own city walls and neighborhoods. These cities live in the same region but don't know each other yet. Time to change that!

VPC Peering (The Friendship Bridge):

Create a magical friendship bridge that lets citizens from one city talk to the other! This peering connection allows smooth travel between cities without using the internet like a secret VIP tunnel.

Update Route Tables (City Maps):

Each city now updates its travel maps to include the new bridge. This means residents know exactly how to reach their new friends on the other side safely and directly.

Connectivity Test (Friendship Test):

Send friendly messages (pings) across the bridge to confirm the connection is working. If everyone waves back, your cities are now best buddies, sharing resources and secrets securely!

BY
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THANK YOU!

